

OD1, the first toxin isolated from the venom of the scorpion *Odonthobuthus doriae* active on voltage-gated Na⁺ channels

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Abstract In this study, we isolated and pharmacologically characterized the first α -like toxin from the venom of the scarcely studied Iranian scorpion *Odonthobuthus doriae*. The toxin was termed OD1 and its primary sequence was determined: GVRDA-YIADDKNCVYTASNGYCNTCTKNGAESGYCQWIGR-YGNACWCIKLPDEVPIRIPGKCR. Using the two-electrode voltage clamp technique, the pharmacological effects of OD1 were studied on three cloned voltage-gated Na⁺ channels expressed in *Xenopus laevis* oocytes (Na_v1.2/β₁, Na_v1.5/β₁, para/tipE). The inactivation process of the insect channel, para/tipE, was severely hampered by 200 nM of OD1 (EC₅₀ = 80 ± 14 nM) while Na_v1.2/β₁ still was not affected at concentrations up to 5 μM. Na_v1.5/β₁ was influenced at micromolar concentrations. © 2005 Federation of European Biochemical Societies. Published by Elsevier B.V. All rights reserved.

Keywords: Iranian scorpion *Odonthobuthus doriae*; α -like toxin; Voltage-gated Na⁺ channel

1. Introduction

Voltage-gated Na⁺ channels (VGSCs) are membrane spanning proteins responsible for action potentials in neurons and most excitable cells [1]. Until this date nine mammalian (Na_v1.1–Na_v1.9) and three insect VGSCs have been cloned [2,3]. Since VGSCs play an important physiological role in vertebrates and invertebrates, they are targeted by toxins from animal venoms and plants [4].

Most scorpion neurotoxins targeting VGSCs are single chain polypeptides composed of 60–70 amino acids cross-linked by 4-disulfide bridges [5–7]. These polypeptides comprise two

main groups: α - and β -toxins [8–10]. Scorpion α -toxins bind to site 3 and prolong the action potential by slowing the inactivation of VGSCs. According to their different pharmacological and binding properties, the α -toxins can be further divided into three subgroups, classical α -, α -like and insect α -toxins. Classical α -toxins (e.g., AaHII, LqhII) are highly toxic to mammals, whereas the insect α -toxins (e.g., Lqh α IT) are highly toxic to insects. The more recently characterized α -like toxins (e.g., BmK M1, LqhIII) act on both mammals and insects, but are unique in their inability to bind to rat synaptosomes despite a high toxicity by intravenous injection [8,11–14].

The Iranian yellow scorpion *Odonthobuthus doriae* is a member of the Buthidae family. The genus Buthus, with the two species *doriae* and *odonturus*, are endemic to the Old World, mainly the Middle East [7,15]. Specifically, the scorpion described here can be found in the central and southern part of Iran. Its sting can cause various effects ranging from local pain, inflammation and necrosis to muscle paralysis and hematuria. Since very little is published about this scorpion and the bioactive substances in its venom, we tried to determine whether VGSC toxins were present.

As a consequence, we report here the isolation and pharmacological characterization of OD1, the first α -like toxin from the venom of the Iranian yellow scorpion *O. doriae* scorpion.

2. Materials and methods

2.1. Purification

500 mg of the crude venom (gift from Dr. Akbary; Vaccine and Serum production and Research Institute of Razi, Karaj, Iran) was dissolved in 10 ml of mobile phase (100 mM ammonium acetate in water, pH 6.8) (Merck, Germany). After sample clarification by centrifugation, fractionation was carried out using two 210 × 3.5 cm Sephadex[®] G₅₀ columns in series (flow rate: 1 ml/min) (Pharmacia, Sweden) at room temperature (20–22 °C) [16,17]. Absorbance was measured at 280 nm. Toxicity guided separation of various protein fractions was accomplished after i.v. injection into mice (white, male, 18–20 g). Fraction 4, containing the toxin, was further purified by ion exchange chromatography (Pharmacia) using a 2.5 × 15 cm column (elution buffers: Tris 50 mM at pH 8 (solution A) and Tris 50 mM at pH 8 plus 1 M NaCl (Solution B)). The first equilibrium step of 60 ml of solution A was followed by a gradient from solution A to solution B (240 ml). The flow rate was 2 ml/min and the absorbance was measured at 280 nm.

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Abbreviations: DTT, dithiothreitol; MS, mass spectrometry; RP-HPLC, reversed-phase high performance liquid chromatography; TEVC, two-electrode voltage clamp; TFA, trifluoro acetic acid; VGSC(s), voltage-gated Na⁺ channel(s)

Lyophilized fraction 3 of the ion exchange chromatography was further purified on a semi-preparative reversed-phase high performance liquid chromatography (RP-HPLC) Vydac® C₈ column (250 × 10 mm, 5 μm) (Hesperia, CA, USA) equilibrated with 0.1% w/v trifluoro acetic acid (TFA). Elution was performed using a linear gradient (20–30%) of acetonitrile containing 0.1% w/v TFA in 45 min (flow rate 3 ml/min). Absorbance was recorded at 215 nm. The final purity of the toxin was assessed on an analytical Supelco® RP-HPLC column (250 × 4.6 mm, 5 μm) using solution A (0.1% TFA in water) and solu-

tion B (0.1% TFA in acetonitrile) with a gradient up till 60% B over a period of 45 min (SMART®, Pharmacia). Flow rate and absorbance were 500 μl/min and 214 nm, respectively.

2.2. Mass spectrometry

The molecular mass of OD1 and fragments generated by enzymatic cleavage were determined by MALDI-TOF mass spectrometry (MS) on a Reflex IV (Bruker Daltonics GmbH, Germany) using a recrystal-

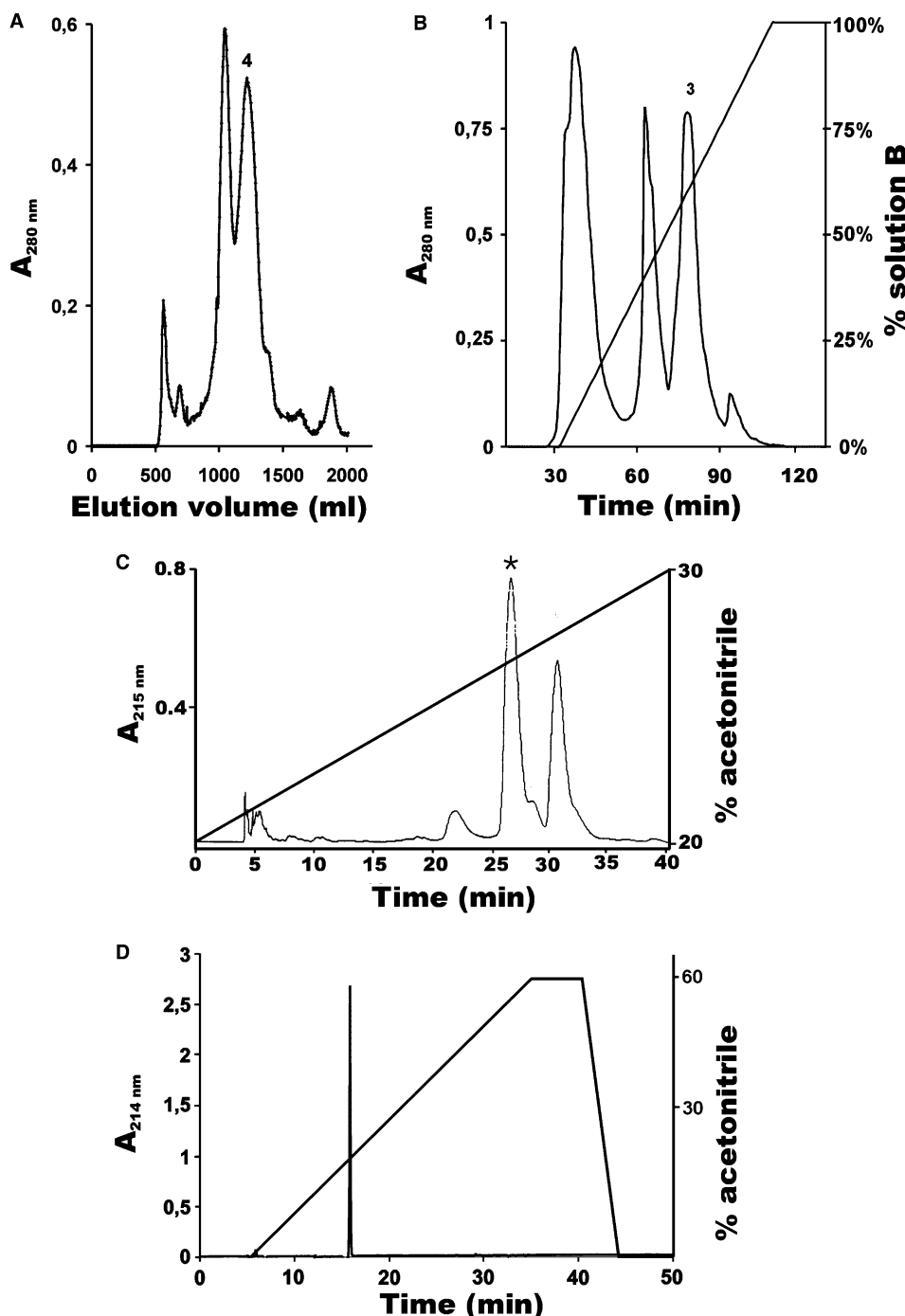


Fig. 1. Purification of OD1 from the venom of *O. doriae*. (A) Crude venom (500 mg) was dialyzed and 107 mg was fractionated by gel filtration. Toxic fraction 4 was recovered and dried. (B) Ion exchange chromatography of fraction 4 (9 mg) obtained in (A). Solution B consists of Tris 50 mM (pH 8) plus 1 M NaCl. (C) Fraction 3 (2 mg) of the ion exchange step was further purified using a C₈ column. Fractions eluting at 26.9 min (*) were recovered and dried (800 μg). (D) The purity of the toxin was assessed on an analytical RP-HPLC C₁₈ column. OD1 encompasses about 1% of the total venom.

lized α -cyano-4-hydroxycinnamic acid matrix. Calibration was performed using standard peptide and protein mixtures (Bruker Daltonics GmbH).

2.3. Sequence determination

The first 47 residues of the primary structure of the OD1 were determined by automated Edman degradation (Procise, Applied Biosystem, USA) (Fig. 2). Nanoflow electrospray ionization hybrid quadrupole time of flight MS was performed on a Q-TOF system (Micromass, UK). For OD1, 1 μ l of a 41.1 μ M OD1 solution and for enzymatic fragments, 1 μ l of the collected fraction was used.

In enzymatic cleavage, 50 μ l of a 41.1 μ M OD1 solution was treated with 1 μ g of modified porcine trypsin (Promega) in 0.2 M NH_4CO_3 (pH 8) for 18 h at 37 °C and subsequently with 50 mM dithiothreitol (DTT) for 2 h at 37 °C. Tryptic fragments were separated on a Waters Symmetry C_{18} (4.6 $\times$ 250 mm, 5 μ M) column using a gradient from 1% to 50% acetonitrile with 0.1% TFA in 40 min (flow rate: 1 ml/min). Peaks were collected manually.

Next, 50 μ l of a 41.1 μ M OD1-solution was treated with 50 mM DTT for 2 h at 37 °C and with 0.2 μ g endoproteinase Asp-N (Roche) in 10 mM Tris (pH 8) for 24 h at 37 °C. Resulting fragments were separated on a Waters Symmetry C_{18} column using a gradient from 1% to 50% acetonitrile with 0.1% TFA in 30 min (flow rate: 1 ml/min). Peaks were collected manually.

2.4. Electrophysiological recordings

For the expression in *Xenopus laevis* oocytes, the $\text{Na}_v1.5$ and β_1 genes were subcloned into pSP64T [18]. For in vitro transcription, $\text{Na}_v1.5$ /pSP64T was first linearized with *Xba*I and β_1 /pSP64T with

*Eco*RI. Next, capped cRNA's were synthesized using the large-scale SP6 mMESSAGE-mACHINE transcription kit (Ambion, USA). The para/pGH19-13-5 vector, tipE/pGH19 vector and $\text{Na}_v1.2$ /pLCT1 vector were linearized with *Not*II and transcribed with the T7 mMESSAGE-mACHINE kit (Ambion) [19–21].

The harvesting of oocytes from anaesthetized female *X. laevis* frogs was as previously described [22]. The ND96 solution was used for incubating the oocytes and contained (in mM): 96 NaCl, 2 KCl, 1.8 CaCl_2 , 1 MgCl_2 , 5 HEPES, pH 7.4, supplemented with 50 mg/l gentamicin sulfate and 180 mg/l theophyllin. Two-electrode voltage clamp (TEVC) recording on oocytes was as previously described [8,11–14]. Whole cell currents were recorded 2–4 days after injection. Currents were filtered at 1 kHz and sampled at 5 kHz. Digital leak subtraction of the current records was carried out using a P/2 protocol. The degree of fast inactivation was assayed by measuring the $I_{10\text{ms}}/I_{\text{peak}}$ ratio which gives an estimate of the probability for the channels not to be inactivated after 10 ms [5]. Curve manipulations were performed using pClamp8 (Axon instruments, USA) and Origin software (Microcal, USA).

3. Results

3.1. Purification and characterization

After purification of the bio-active fraction using conventional chromatography techniques (as shown in Fig. 1) sequence determination of the purified toxin yielded a peptide of 65 residues, amidated at the C-terminus, termed OD1.

A	1. GVRDAYIADDKNCVYTCASNGYCNTECTKNGAESGYCQWIGRYGNAC										
	2. YGNACWCNIK										
	3. LPDEVPIR										
	4. DEVPIRIPGKCRamide										
B											
	10	20	30	40	50	60					
OD1	GVRDAYIADDKNCVYTCAS-NGYCNTECTKNGAESGYCQWIGRYGNACWCNIKLPDEVPIRIPG-KCR										%100
	BBBBB	TTTTT	HHHHHHHHH	BBBBB	BBBBBBB						
alpha-like toxins											
BmKM1	-VRDAYIAKPHNCVYECAR-NEYCNLDCTKNGAKSGYCWQWVGKYGNGCWCIELPDNVPIRVPG-KCH--										75
BmKM2	-VRDAYIAKPHNCVYECAR-NEYCNLDCTKNGAKSGYCWQWVGKYGNGCWCIELPDNVPIRVPG-KCH--										75
BomIV	-GRDAYIAQPENCVYECAR-NSYCNLDCTKNGAKSGYCWQWVGKYGNGCWCIEDLPDNVPIRIPG-KCHF--										73
BmKM7	-VRDGYIALPHNCAYGCLN-NEYCNLDCTKNGAKSGYCWQWVGKYGNGCWCIELPDNVPIRVPG-KCHPA										63
Lqh3	-VRDGYIAQPENCVYHCFPGSSGCDTLCCKEGGTSGHCGFKVGHGLACWCNLPDNVGIIVDGVKCH--										44
BomIII	-GRDGYIAQPENCVYHCFPGSSGCDTLCCKEGGTSGHCGFKVGHGLACWCNLPDNVGIIVDGVKCH--										44
Lqh7	-VRDGYIAKPENCAHHCFPGSSGCDTLCCKEGGTSGHCGFKVGHGLACWCNLPDNVGIIVDGVKCH--										41
alpha-toxins											
LqqIV	-VRDAYIADDKNCVYTCGS-NSYCNTECTKNGAESGYCWQWVGKYGNGCWCNIKLPDKVPIRIPG-KCR--										92
LqhIV	-VRDAYIADDKNCVYTCGA-NSYCNTECTKNGAESGYCWQWVGKYGNGCWCNIKLPDKVPIRIPG-KCR--										91
BotI	-GRDAYIAQPENCVYECAR-NSYCNLDCTKNGATSGYCWQWVGKYGNGCWCNKDLPDNVPIRIPG-KCHF--										73
BotII	-GRDAYIAQPENCVYECAR-NSYCNLDCTKNGAKSGYCWQWVGKYGNGCWCNKDLPDKVPIRIPG-KCHF--										72
AaHII	-VKDGYIVDDVNCYFCGR-NAYCNEECTKLKGESGYCWQWVGKYGNGCWCNKDLPDKVPIRIPG-KCHF--										60
LqqV	-LKDGYIVDDKNCYFCGR-NAYCNDECKKKGGESGYCWQWVGKYGNGCWCNKDLPDKVPIRIPG-KCHS--										60
BotIII	-VKDGYIVDDKNCYFCGR-NAYCNEECTKLKGESGYCWQWVGKYGNGCWCNKDLPDKVPIRIPG-KCHF--										59
Insect alpha-toxins											
BjalphaIT	-GRDAYIADNLNCAAYTCGS-NSYCNTECTKNGAVSGYCWQWVGKYGNGCWCNIKLPDKVPIRIPG-ACR--										79
LqhalphaIT	-VRDAYIAKYNVCVYECFR-DAYCNELCTKNGASSGYCWQWVGKYGNGCWCNKDLPDNVPIRVPG-KCH--										73
BotIT1	-VRDAYIAQYNVCVYECMK-DDYCNLDCTKNGASSGYCWQWVGKYGNGCWCNKDLPDNVPIRIPG-KCHS--										72
LqqIII	-VRDAYIAKYNVCVYECFR-DSYCNLDCTKNGASSGYCWQWVGKYGNGCWCNKDLPDNVPIRVPG-KCH--										71
	*	*	*	*	*	*	*	*	*	*	

Fig. 2. (A) (1) Result from direct Edman degradation of the entire toxin. (2) and (3) Trypsin cleavage followed by MALDI-TOF MS and Q-TOF MS (confirmatory). Edman degradation was used for distinguishing between leucine/isoleucine and glutamine/lysine. (4) AspN cleavage followed by MALDI-TOF MS and Q-TOF MS (confirmatory). Edman degradation was used for distinguishing between leucine/isoleucine and glutamine/lysine. (B) Comparison of the amino-acid sequence of OD1 with other similar toxins. Secondary structures are indicated on top (B, β -sheet; T, turn; H, α -helix). (*) Residues conserved in all listed toxins. The C-terminus of OD1 is amidated. Sequences taken from [6,7,29].

Spectrometric measuring of the toxin resulted in a mass of 7204.8 Da (theoretical mass: 7206.2 Da).

The sequence of OD1 was aligned with other similar sequences by using ClustalW (<http://www.ebi.ac.uk/clustalw>). Scorpion α -toxins with high similarity include LqqIV (from *Leiurus quinquestriatus quinquestriatus*, 92%), LqhIV (from *Leiurus quinquestriatus hebraeus*, 91%), BjaIT (from *Buthus judaicus*, 79%) and BmK M1/BmK M2 (from *Buthus martensii* Karsch, 75%) (see Fig. 2) [6,14,23].

3.2. Effects of OD1 on VGSCs

Fig. 3 shows the maximum efficacy of OD1 on $\text{Na}_v1.2/\beta_1$ (5 μM), $\text{Na}_v1.5/\beta_1$ (2 μM) and the insect VGSC para/tipE (200 nM). Current traces were the result of a TEVC experiment on VGSCs expressed together with their β_1 subunit (tipE for para). Current–voltage relationships were constructed by using 50 ms step depolarizations to a voltage range between -70 and 40 mV from a holding potential of -90 mV. Current traces recorded after the addition of OD1 reveal a slowing of

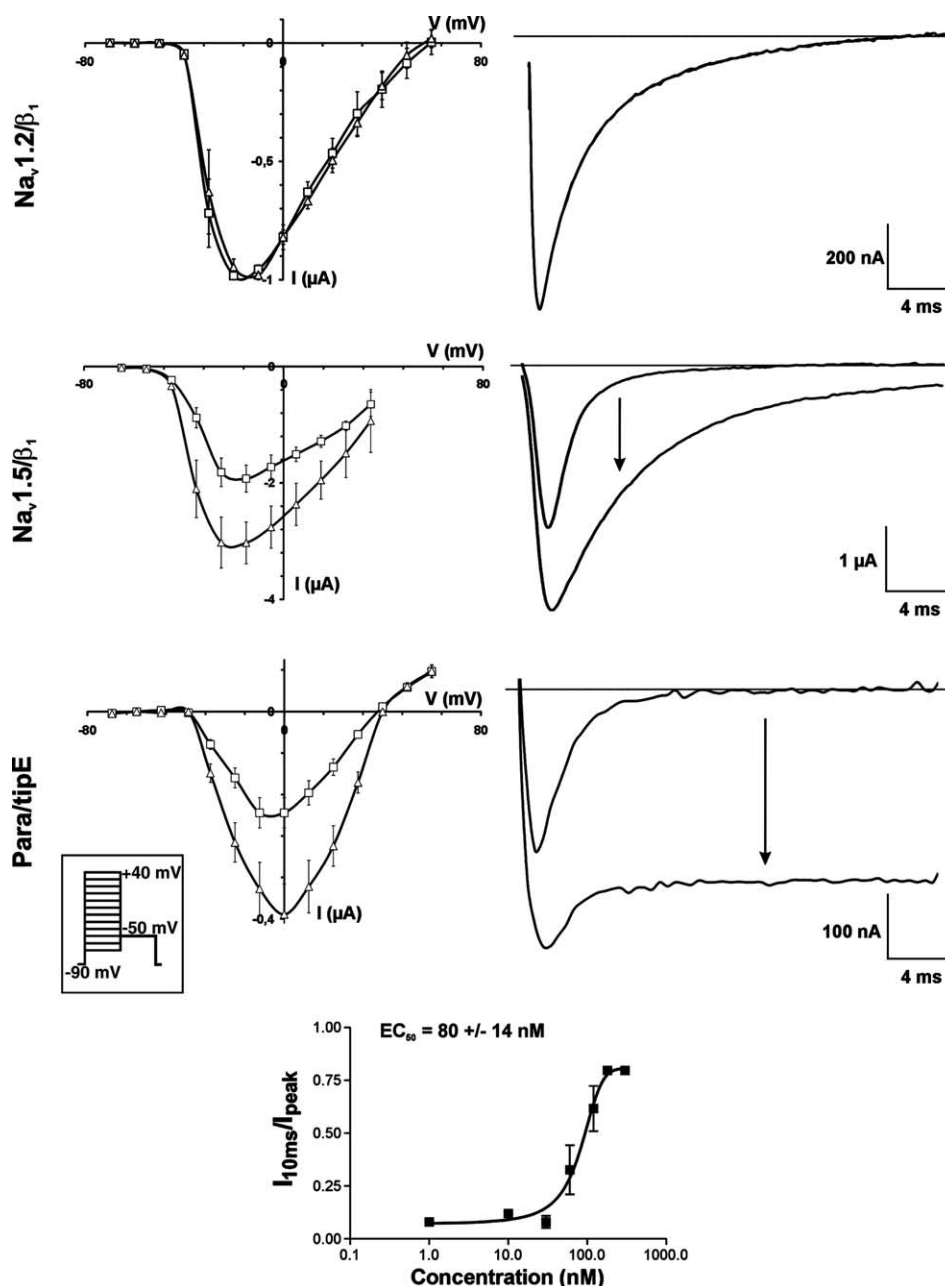


Fig. 3. Effect of OD1 on the inactivation kinetics of $\text{Na}_v1.2/\beta_1$, $\text{Na}_v1.5/\beta_1$ and para/tipE expressed in *X. laevis* oocytes. Concentrations shown are: $\text{Na}_v1.2/\beta_1$, 5 μM ; $\text{Na}_v1.5/\beta_1$, 2 μM ; para/tipE, 200 nM. No effect was observed after addition of OD1 (2 μM) on $\text{Na}_v1.2/\beta_1$. An obvious slowing of inactivation was observed for the other VGSCs. The voltage protocol with which the I – V curves were obtained is shown at the lower left corner. I – V curves reveal a large increase of Na^+ ions influx after addition of OD1 (Δ) in comparison to control conditions ($\square$). No shift in activation voltage or reversal potential was seen. Note that the initial fast phase of activation was not affected by the toxin, the slowing of the late phase of inactivation and the increasing peak amplitude. Dose–response curve of OD1 on para/tipE ($\blacksquare$) is shown at the bottom. Data are means $\pm$ S.E. mean of at least three experiments. A Boltzmann sigmoidal fit of the $I_{10\text{ms}}/I_{\text{peak}}$ ratio data (see Section 2) provided the EC_{50} value of 80 ± 14 nM.

the inactivation process of $\text{Na}_v1.5/\beta_1$ and para/tipE. Fig. 3 reveals that OD1 does not affect $\text{Na}_v1.2/\beta_1$ at concentrations up to 5 μM . On $\text{Na}_v1.5/\beta_1$, the slowing of inactivation only became visible at concentrations above 1 μM . Para/tipE is by far the most sensitive channel since OD1 completely inhibits the inactivation process when 200 nM is added. A shift neither in activation nor in reversal potential is seen on the studied VGSCs. The slowing of inactivation induced by OD1 is concentration-dependent. The EC_{50} value of OD1 on $\text{Na}_v1.5/\beta_1$ was not determined since OD1 only had an effect on this VGSC at high concentrations (2 μM). The EC_{50} value of OD1 on para/tipE was determined with a Boltzmann sigmoidal fit of the obtained data and was 80 ± 14 nM.

4. Discussion

For the first time, an α -like toxin from the venom of the Iranian Yellow scorpion *O. doriae*, OD1, was purified and pharmacologically characterized on three cloned VGSCs. OD1 does not influence the inactivation process of $\text{Na}_v1.2/\beta_1$. This toxin does hamper the inactivation of $\text{Na}_v1.5/\beta_1$ but only at high concentrations (2 μM). The main target is clearly the insect VGSC, para/tipE, of which the inactivation process is completely inhibited when 200 nM of OD1 is added ($\text{EC}_{50} = 80 \pm 14$ nM). The original sub-classification of scorpion α -toxins relies on competition experiments and relative toxicity tests [9]. Since we report here the effects of OD1 on three cloned VGSCs, the obtained data are compared to what is known for other α -like toxins, electrophysiologically tested on cloned VGSCs [24,25]. For instance, the activity of the α -like toxin BmK M1 has been characterized on $\text{Na}_v1.2/\beta_1$, $\text{Na}_v1.5/\beta_1$ and para/tipE and is similar to OD1 [14]. Furthermore, the pharmacological profile of LqhIII, also regarded as an α -like toxin [5,24,25], and OD1 (44% similarity) are alike. Indeed, the fact that OD1 acts on VGSCs of both mammals and insects is in concordance with the definition of α -like toxins derived from binding studies. Therefore, based on its trademarks, OD1 can be regarded as a scorpion α -like toxin but with a distinct preference for the insect VGSC.

Although the primary structure of OD1 possesses 92% similarity with the α -toxin LqqIV [23] (targeting mammalian VGSCs) (see Fig. 2), our bioassay reveals a distinct preference for the insect VGSC, para. The region consisting of the reverse turn (residues 8–12) in combination with C-tail (57–64), has received much attention because of its involvement in the binding preference for phylogenetically distinct VGSCs [26,27]. A recent study from Karbat et al. [28] reported the transfer of these two motifs (termed NC-domain) of a scorpion insect α -toxin (Lqh α IT) to the scaffold of the anti-mammalian scorpion α -toxin AaH2 resulting in activity on insects. Furthermore, previous studies on BmK M1 revealed three functionally important epitopes that underlie the molecular basis of mammal and/or insect-selectivity: (1) the first three N-terminal residues; (2) the five-residue reverse turn (residues 8–12) in combination with the C-tail (residues 57–64) and (3) the loop between the β_1 and β_2 sheet (residues 40–43) [14]. Comparing the sequence of OD1 to that of BmK M1, we see that the region consisting of the reverse turn is remarkably different between these two toxins. While K8/P9/H10 in BmK M1 is mostly basic, D8/D9/K10 in OD1 is more acidic. In addition, α -like toxins possess a proline at position 9 while most α -toxins

do not. Since a proline causes a nick in the secondary structure, the absence of this residue in OD1 can cause a different structure at this important region. Remarkably, LqqIV possesses the same sequence in this region as OD1. Even the C-tail of these two toxins is identical. Furthermore, a recent study from Arnon et al. [6] describes a novel toxin, Bj α IT, as being an insect-selective scorpion α -toxin which also shares high similarity with LqqIV. However, based on LD_{50} values on mice and competition studies on rat brain synaptosomes, LqqIV has been classified as an α -toxin [23], in contrast to Bj α IT and OD1. Nonetheless, Cestèle et al. [13] reported that LqqIV does possess a low toxicity in insects and that this toxin is able to bind to cockroach neuronal membranes [23]. Therefore, LqqIV can in fact be considered an α -like toxin (like OD1) but with a distinct preference for mammals (unlike OD1, which has a distinct preference for insects).

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References

- [1] Catterall, W.A. (2000) From ionic currents to molecular mechanisms: the structure and function of voltage-gated sodium channels. *Neuron* 26, 13–25.
- [2] Yu, F.H. and Catterall, W.A. (2003) Overview of the voltage-gated sodium channel family. *Genome Biol.* 4, 207.
- [3] Zlotkin, E. (1999) The insect voltage-gated sodium channel as target of insecticides. *Annu. Rev. Entomol.* 44, 429–455.
- [4] Wang, S.Y. and Wang, G.K. (2003) Voltage-gated sodium channels as primary targets of diverse lipid-soluble neurotoxins. *Cell Signal.* 15, 151–159.
- [5] Chen, H., Lu, S., Leipold, E., Gordon, D., Hansel, A. and Heinemann, S.H. (2002) Differential sensitivity of sodium channels from the central and peripheral nervous system to the scorpion toxins Lqh-2 and Lqh-3. *Eur. J. Neurosci.* 16, 767–770.
- [6] Arnon, T., Potikha, T., Sher, D., Elazar, M., Mao, W., Tal, T., Bosmans, F., Tytgat, J., Ben-Arie, N. and Zlotkin, E. (2005) Bj α IT: a novel scorpion α -toxin selective for insects—unique pharmacological tool. *Insect Biochem. Mol. Biol.* 35, 187–195.
- [7] Possani, L.D., Becerril, B., Delepierre, M. and Tytgat, J. (1999) Scorpion toxins specific for Na^+ -channels. *Eur. J. Biochem.* 264, 287–300.
- [8] Gordon, D., Martin-Eauclaire, M.F., Cestèle, S., Kopeyan, C., Carlier, E., Khalifa, R.B., Pelhate, M. and Rochat, H. (1996) Scorpion toxins affecting sodium current inactivation bind to distinct homologous receptor sites on rat brain and insect sodium channels. *J. Biol. Chem.* 271, 8034–8045.
- [9] Gordon, D., Savarin, P., Gurevitz, M. and Zinn-Justin, S. (1998) Functional anatomy of scorpion toxins affecting sodium channels. *J. Toxicol. Toxin Rev.* 17, 131–159.
- [10] Leipold, E., Lu, S., Gordon, D., Hansel, A. and Heinemann, S.H. (2004) Combinatorial interaction of scorpion toxins Lqh-2, Lqh-3, and Lqh α IT with sodium channel receptor sites-3. *Mol. Pharmacol.* 65, 685–691.
- [11] Gilles, N., Blanchet, C., Shichor, I., Zaninetti, M., Lotan, I., Bertrand, D. and Gordon, D. (1999) A scorpion α -like toxin that is active on insects and mammals reveals an unexpected specificity and distribution of sodium channel subtypes in rat brain neurons. *J. Neurosci.* 19, 8730–8739.
- [12] Sautiere, P., Cestèle, S., Kopeyan, C., Martinage, A., Drobecq, H., Doljansky, Y. and Gordon, D. (1998) New toxins acting on sodium channels from the scorpion *Leiurus quinquestriatus*

- hebraeus* suggest a clue to mammalian vs insect selectivity. *Toxicon* 36, 1141–1154.
- [13] Cestèle, S., Stankiewicz, M., Mansuelle, P., De Waard, M., Dargent, B., Gilles, N., Pelhate, M., Rochat, H., Martin-Eauclaire, M.F. and Gordon, D. (1999) Scorpion alpha-like toxins, toxic to both mammals and insects, differentially interact with receptor site 3 on voltage-gated sodium channels in mammals and insects. *Eur. J. Neurosci.* 11, 975–985.
 - [14] Liu, L.H., Bosmans, F., Maertens, C., Zhu, R.H., Wang, D.C. and Tytgat, J. (2005) Molecular basis of the mammalian potency of the scorpion alpha-like toxin, BmK M1. *FASEB J.* 19, 594–596.
 - [15] Latifi, M. and Tabatabai, M. (1979) Immunological studies on Iranian scorpion venom and antiserum. *Toxicon* 17, 617–621.
 - [16] D'Suze, G., Corona, F., Possani, L.D. and Sevcik, C. (1996) High performance liquid chromatography purification and amino acid sequence of toxins from the muscarinic fraction of *Tityus discrepans* scorpion venom. *Toxicon* 34, 591–598.
 - [17] Amininasab, M., Elmi, M.M., Endlich, N., Endlich, K., Parekh, N., Naderi-Manesh, H., Schaller, J., Mostafavi, H., Sattler, M., Sarbolouki, M.N. and Muhle-Goll, C. (2004) Functional and structural characterization of a novel member of the natriuretic family of peptides from the venom of *Pseudocerastes persicus*. *FEBS Lett.* 557, 104–108.
 - [18] Gellens, M.E., George Jr., A.L., Chen, L.Q., Chahine, M., Horn, R., Barchi, R.L. and Kallen, R.G. (1992) Primary structure and functional expression of the human cardiac tetrodotoxin-insensitive voltage-dependent sodium channel. *Proc. Natl. Acad. Sci. USA* 89, 554–558.
 - [19] Warmke, J.W., Reenan, R.A., Wang, P., Qian, S., Arena, J.P., Wang, J., Wunderler, D., Liu, K., Kaczorowski, G.J., Van der Ploeg, L.H., Ganetzky, B. and Cohen, C.J. (1997) Functional expression of *Drosophila* para sodium channels. Modulation by the membrane protein TipE and toxin pharmacology. *J. Gen. Physiol.* 110, 119–133.
 - [20] Feng, G., Deak, P., Chopra, M. and Hall, L.M. (1995) Cloning and functional analysis of TipE, a novel membrane protein that enhances *Drosophila* para sodium channel function. *Cell* 82, 1001–1011.
 - [21] Noda, M., Ikeda, T., Kayano, T., Suzuki, H., Takeshima, H., Kurasaki, M., Takahashi, H. and Numa, S. (1986) Existence of distinct sodium channel messenger RNAs in rat brain. *Nature* 320, 188–192.
 - [22] Liman, E.R., Tytgat, J. and Hess, P. (1992) Subunit stoichiometry of a mammalian K⁺ channel determined by construction of multimeric cDNAs. *Neuron* 9, 861–871.
 - [23] Kopeyan, C., Martinez, G. and Rochat, H. (1985) Primary structure of toxin IV of *Leiurus quinquestriatus quinquestriatus*. *FEBS Lett.* 181, 211–217.
 - [24] Chen, H., Gordon, D. and Heinemann, S.H. (2000) Modulation of cloned skeletal muscle sodium channels by the scorpion toxins LqhII, LqhIII, and LqhalphalIT. *Pflügers Arch.* 439, 423–432.
 - [25] Chen, H. and Heinemann, S.H. (2001) Interaction of scorpion alpha-toxins with cardiac sodium channels: binding properties and enhancement of slow inactivation. *J. Gen. Physiol.* 117, 505–518.
 - [26] Guan, R.J., Xiang, Y., He, X.L., Wang, C.G., Wang, M., Zhang, Y., Sundberg, E.J. and Wang, D.C. (2004) Structural mechanism governing *cis* and *trans* isomeric states and an intramolecular switch for *cis/trans* isomerization of a non-proline peptide bond observed in crystal structures of scorpion toxins. *J. Mol. Biol.* 341, 1189–1204.
 - [27] He, X.L., Li, H.M., Zeng, Z.H., Liu, X.Q., Wang, M. and Wang, D.C. (1999) Crystal structures of two alpha-like scorpion toxins: non-proline *cis* peptide bonds and implications for new binding site selectivity on the sodium channel. *J. Mol. Biol.* 292, 125–135.
 - [28] Karbat, I., Frolow, F., Froy, O., Gilles, N., Cohen, L., Turkov, M., Gordon, D. and Gurevitz, M. (2004) Molecular basis of the high insecticidal potency of scorpion alpha-toxins. *J. Biol. Chem.* 279, 31679–31686.
 - [29] Hamon, A., Gilles, N., Sautiere, P., Martinage, A., Kopeyan, C., Ulens, C., Tytgat, J., Lancelin, J.M. and Gordon, D. (2002) Characterization of scorpion alpha-like toxin group using two new toxins from the scorpion *Leiurus quinquestriatus hebraeus*. *Eur. J. Biochem.* 269 (16), 3920–3933.